

Multiple Choice (10 marks)

1. Which of the following statements is not true?
 - (a) Docosahexaenoic acid can be synthesised from linolenic acid in vegans
 - (b) Docosahexaenoic acid is absent from vegan diets
 - (c) Lack of docosahexaenoic acid in vegans causes visual and cognitive impairment
 - (d) Microalgae can be used to synthesise docosahexaenoic acid
2. According to the recommendations for the nutritional management of patients with diabetes, the consumption of saturated fat should be:
 - (a) <10% of total daily energy
 - (b) <12%
 - (c) <15%
 - (d) <16%
3. Which of the following changes in body composition are associated with ageing in older adults?
 - (a) Increased bone mass
 - (b) Increased muscle mass
 - (c) Increased body fat
 - (d) All of the above
4. What are the limitations of food balance sheets? (select all that apply)?
 - (a) No account is taken of waste or losses of food
 - (b) Consumption estimates give higher values than those derived from other types of surveys
 - (c) No information is available on the distribution of consumption within a country
 - (d) All of the options given are correct
5. Which of the factors below is NOT considered a risk factor for Eating Disorders
 - (a) Perfectionism traits
 - (b) Female gender
 - (c) Dieting during adolescence
 - (d) Parental anxiety

True / False (10 marks)

Answer True or False

6. True or False: Epigenetic changes can include DNA methylation, modify gene expression, and lead to different phenotypes from the same genotype.
7. True or False: Proline, tyrosine, and serine are essential amino acids that need to be consumed through food.
8. True or False: Protein-energy supplementation during pregnancy and the first two years can improve child development.
9. True or False: Vitamin D, vitamin C, and vitamin A are essential for reducing homocysteine levels in the bloodstream.
10. True or False: The WHO recommends that individuals should limit their intake of free sugars to less than 10% of their total energy intake.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the signs and symptoms that indicate an infant may have a severe combined immunodeficiency. How do these signs reflect the underlying issues with the immune system?
12. Discuss the role of different types of carbohydrates in oral health, specifically focusing on why starch cannot be metabolized by oral bacteria compared to other carbohydrates.

End of Paper